LUCKY SAH

Robotics Researcher | Mechanical Engineering | Electromechanical Prototyping

Kent, OH | lsah@kent.edu | *[ADD PHONE]* | *[ADD LINKEDIN OR PORTFOLIO URL]*

PROFESSIONAL SUMMARY

Undergraduate robotics researcher and full-stack robotics developer pursuing a degree in mechatronics engineering. Develops humanoid robotic systems from mechanical design and fabrication through hardware integration, robotics software, and simulation, using SolidWorks, Autodesk Inventor, ROS, Isaac Sim, and MuJoCo. Coauthored research papers address mechanical systems, agentic robotics, and human–robot interaction.

RESEARCH & ENGINEERING EXPERIENCE

Robotics Researcher *[ADD START DATE] – Present*

Advanced Telerobotics Research (ATR) Lab, Kent State University

- Lead mechanical development of mobile robots and humanoid robotic mechanisms, designing structures, actuator mounts, and sensor assemblies from CAD through working prototypes.
- Develop and integrate robotics software in ROS; build simulation models in Isaac Sim and MuJoCo to evaluate robot behavior and test hardware–software integration.
- Integrate actuators, sensors, and embedded electronics; diagnose mechanical and control issues during prototype testing.

Robotics Hardware Intern (Co-op) *[ADD DATES]*

PlaiPin Inc.

- Lead mechanical CAD design for a modular AI companion robot, developing the core device and interchangeable body configurations around actuators, sensors, and onboard electronics.
- Build and refine functional prototypes, resolving assembly fit and component-packaging issues through iterative fabrication and hardware integration.

Automation and Product Development Intern *2022–2024*

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- Designed, fabricated, assembled, and tested mechanical systems for automated production machinery.
- Assembled industrial equipment, supported PLC wiring and lab setup, and fabricated parts using machining and fiber-reinforced molding.

SELECTED PUBLICATIONS

I. S. Cardenas, M. A. Arnett, N. C. Yeo, **L. Sah**, and J.-H. Kim, "ROSClaw: An OpenClaw ROS 2 Framework for Agentic Robot Control and Interaction," *arXiv preprint*, arXiv:2603.26997, 2026.

M. Chen, **L. Sah**, A. Rivera, and D. Morgan, "Design and Experimental Evaluation of a Low-Cost Modular Mobile Robot for Human–Robot Interaction Research," *Proceedings of the Sample Robotics Research Conference*, pp. 112–119, 2026. *(fictional layout sample)*.

A. Rivera, M. Chen, **L. Sah**, and D. Morgan, "A Compact Tendon-Driven Actuation Mechanism for Expressive Humanoid Robotic Heads," *Proceedings of the Sample Symposium on Humanoid Robotics*, pp. 48–55, 2026. *(fictional layout sample)*.

SELECTED ENGINEERING PROJECTS

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed a chassis and motion architecture around low-cost fabrication, modular expansion, and straightforward integration of robotic hardware. *[ADD TEST RESULT OR COST]*

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head integrating mechanical motion systems and embedded components for human–robot interaction; project resulted in a **CoRL 2025 workshop poster paper**.

TECHNICAL SKILLS

Mechanical Design: Autodesk Inventor Professional, Fusion 360, SolidWorks, parametric assemblies, mechanisms, 3D modeling, bill-of-materials development

Robotics & Embedded: ROS, Arduino, microcontrollers, actuators, sensors, motor drivers, electromechanical integration

Fabrication: FDM and resin 3D printing, CNC machining, soldering, power tools, silicone, polyurethane, and fiber-reinforced molding

Automation: PLC fundamentals, industrial equipment assembly, wiring, and production-oriented fabrication

EDUCATION

B.S. in Mechatronics Engineering; Minor in Computer Science *Expected 2028*

Kent State University — Kent, Ohio

INTERNATIONAL ACTIVITIES

Selected competitions & challenges: World Robot Olympiad (Thailand); FIRST Global Challenge (Switzerland); Asian Development Bank Hackathon; Robo-Battle Bots; Robo Soccer. *[ADD ROLE, YEAR, AND RESULT]*

Selected conferences & programs: Founders, Inc. (San Francisco); Asian Pacific Conference of Young Scientists. *[ADD ROLE]*